THE STRUCTURE OF MATHEMATICAL REALITY

Phase III — Chapter 9

Topology: Structure Without Measurement

Note Template

Self-Study Curriculum

How to use this template

Complete these notes **after** the reading and the lecture. The goal is not to copy phrases from the text — it is to reconstruct the ideas in your own language. Leave every item blank on first pass; fill it in from memory, then check.

1. What topology studies

What it preserves:

What it throws away:

2. Why measurement-free structure matters

give a reason from the lecture that resonated with you

3. The fundamental concept in topology is the __________, not

the ____________ or the ____________.

4. Continuity in topology means

restate without using "epsilon-delta"

5. Key concepts

Homeomorphism:
Connectedness:
Compactness:
Fundamental group:

6. The donut vs. sphere distinction captures

what structural difference, exactly?

7. Where topology appears in my own work (implicitly)

parameter spaces? state spaces? fitness landscapes?

8. TDA

What it does:

Why topological invariants are valuable for data: